



BI Requirements

Digital Wellness for Tomorrow

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EXECUTIVE SUMMARY

Project Overview

The Digital Wellness for Tomorrow initiative is a strategic Business Intelligence project led by I.Q. Productionz. It is designed to smooth the peak of negative economic and psychological impacts of excessive screen time and simultaneously strengthen the positive ones. By integrating data from school surveys, wellness applications, HR systems, and the CDC, we will identify behavioral risk patterns in children and employees to provide data-driven, actionable interventions. Our goal is to boost workforce productivity by an estimated 15% while fostering high-value partnerships within the EdTech sector through predictive models and analytics capabilities.

Strategic Assumptions

This project is built on the assumption that screen-time data from external wellness apps and school surveys can be successfully standardized. Additionally, we believe that stakeholders (HR, parents, and educators) will remain actively engaged throughout the project lifecycle. The main phase that we are assuming these factors to be solid in is the intervention phase. We further assume that our current cloud-native infrastructure can scale to handle real-time data feeds while maintaining high data integrity and security compliance.

Risks and Issues

Within the inconsistency of self-reported screen-time metrics across different sources, our primary concern is **Data Quality Risk**. We must also navigate **Interpretation Risk**. When checking the risks within interpretation, we are checking for data correlations that are not misconstrued as absolute causation. As of now, the project faces a challenge regarding the limited availability of longitudinal datasets directly linking early screen habits to long-term employment outcomes, which will be addressed through iterative predictive modeling and ongoing data acquisition efforts.

Key Project Metrics

- **Timeline:** 18 weeks (approximately 4.5 months)
- **Budget:** \$57,000 (Total Project Cost)
- **Total Cost of Ownership (TCO):** Includes ongoing charges. Typically those are renewed annually and consist of platform maintenance, subscriptions and data storage fees.
- **ROI:** Strongly positive, driven by a reduction in mental-health-related absenteeism, elimination of manual reporting through automated BI dashboards, early intervention cost savings, and new EdTech partnership revenue streams

Recommendation

Based on the high strategic coherence with future workforce readiness and the significant potential for revenue growth through wellness innovation, it is the strong recommendation of the project team to proceed with the immediate initiation of Phase 1. The qualitative cost of inaction includes:

1. Potential brand backlash
2. Declining productivity from unidentified at-risk populations
3. Missed partnership opportunities

To eliminate these factors, the projected \$57,000 investment will support the completion of this project.

Project Description

Screen time analytics harnessed will drive business to grow with innovative digital wellness solutions. Risks will be mitigated by identifying at risk groups to recommend interventions to. After usage patterns are analyzed, edtech partnerships are able to deliver actionable insights that boost workforce productivity. Revenue streams are increased and mental health improved by using dashboards that provide predictive models. Offering safe screen time limitation suggestions enhance brand representation mitigating further sales.

Key Deliverables

- **Strategic Data Integration & Methodology**
 - **Strategic Data Integration & Methodology**
 - **Multi-Source Ingestion:** To integrate diverse datasets that include CDC research, real-time school surveys, and wellness app analytics.
 - **Data Integrity:** To focus on continuous data cleansing and up-keep to ensure all interventions are based on high-quality, "fresh" information.
 - **Validation:** To prioritize the ongoing reassessment of predictive models to maintain accuracy in risk identification.
- **Operational Capabilities & BI Delivery**
 - **Unified Environment:** Consolidates fragmented data into a single platform to improve accessibility, governance, and quality assurance.
 - **Advanced Analytics Stack:** Implements three distinct analytical styles:
 - **Descriptive:** To understand current patterns.
 - **Predictive:** To forecast emerging trends and risks.
 - **Prescriptive:** To recommend specific wellness interventions.
 - **Self-Service Empowerment:** To provide interactive, real-time dashboards that allow non-technical stakeholders (HR, educators, parents) to generate reports without relying on IT.

- **Business Impact & Organizational Value**

- **Proactive Wellness:** To enable risk identification as early as possible and personalized digital wellness interventions for at-risk groups.
- **Revenue & Partnerships:** To support product development and foster partnerships with EdTech providers. This supports the goal of opening new revenue streams.
- **Operational Efficiency:** To eliminate manual reporting. This will significantly reduce administrative workload and streamlining departmental workflows.
- **Organizational Agility:** To Facilitate faster, data-driven decision-making. This quick adaptability to market changes and brand reputation risks creates the opportunity for the company to level-up.

Critical Success Factors

- Dependencies
 1. **Timely Access to Quality Data** - Reliable, complete, and consistently-measured screen-time datasets from schools, youth programs, and digital wellness organizations delivered on schedule
 2. **Secure Technical Infrastructure** - IT-provided data storage and processing systems capable of handling sensitive information and integrating multiple data streams
 3. **Active Stakeholder Engagement** - Continuous involvement from sponsors and partners providing guidance and feedback throughout the project lifecycle (formalized through signed agreements)
 4. **Research Partner Compliance** - Behavioral and academic data delivered according to agreed-upon timelines
 5. **Functional Analytical Tools** - Tableau and analytical software platforms operating effectively to visualize complex patterns and process findings
 6. **In-House Expertise** - Sufficient internal knowledge to analyze intersections between academic performance, behavioral patterns, and employment-related activities
 7. **Proactive Risk Identification** - Early warning systems that the team utilizes in order to respond quickly to challenges prior to adverse effects

- Prerequisites

1. **Legal/Compliance:**

- *Data sharing agreements* signed with schools, youth programs, and digital wellness organizations
- *IRB approval* (Institutional Review Board) if working with minors or sensitive data
- *Privacy compliance* (FERPA for educational data, COPPA for children's data and HIPAA if health data is needed)
- *Data anonymization protocols* established and approved

2. **Technical:**

- *Secure infrastructure* already set up (can't start collecting sensitive data without it)
- *Tableau licenses* purchased and installed
- *Cloud storage* provisioned with appropriate security configurations
- *IT support commitment* secured

3. **Organizational:**

- *Budget approved* (\$57,000)
- *Team assembled* (Data Analyst, BI Developer, Project Manager, SME hired or assigned)
- *Stakeholder buy-in* from sponsors (not just "interested" but committed)
- *Signed stakeholder engagement agreement* (as you mentioned drafting)

4. **Data:**

- *Historical screen-time data exists* (longitudinal data can only be created if it was collected)
- *Data accessibility* (partners need to be willing and able to share it)
- *Baseline data quality assessment* to know if retrieving the data is worth the effort.

5. Skills/Knowledge:

- **Skills/Knowledge:**
- *Required expertise* the team must have a solid background with BI development, behavioral research and data analysis
- *Understanding of employment outcome metrics* (what constitutes "employment readiness"?)

Risks

1. Data Quality

- Incomplete/inconsistent data weakens conclusions.
- To provide rigorous validation at every processing stage, inspect for:
 - Missing fields
 - Irregular timestamps
 - Conflicting metrics

2. Stakeholder Misalignment

- Sponsor priority-shift warning signs:
 - Fewer meetings
 - Delayed approvals
 - Vague feedback
- Combat with regular communication and transparent milestone reviews.

3. Interpretation Risk

- Misinterpretations can be redirected by providing statistical disclaimers and clear documentation.
- Findings may be perceived as proving screen time *causes* unemployment issues when our intention is solely to show patterns.

4. Technical Risk

- Mitigating tool pitfalls can be done by:
 - Testing early
 - Having backup tools ready
 - Coordinating closely with IT

Concerns

1. **Limited longitudinal data** directly linking screen time to job outcomes (existing datasets weren't built for this)
2. **Self-reported screen time is unreliable** (everyone measures differently)
3. **Correlation vs. causation confusion** will happen despite our warnings
4. **Political/ideological resistance** from people with strong opinions about screen time concerns

BUSINESS REQUIREMENTS

Table 1: List of Business Requirements

Identifier	Name	Description	Priority
BR-01	Screen Time Risk Identification	Ability to identify at-risk groups (children, employees, students) based on screen-time patterns and behavioral indicators.	H
BR-02	Productivity Impact Analysis	Quantify how screen time correlates with employee productivity, absenteeism, and wellness outcomes.	H
BR-03	Digital Wellness Intervention Recommendations	Provide data-driven recommendations for screen-time reduction and wellness interventions.	H
BR-04	EdTech & Wellness Partnership Enablement	Supply analytics that support partnerships with EdTech and wellness solution providers.	M
BR-05	Mental Health Trend Monitoring	Track mental-health-related indicators linked to excessive screen time.	H
BR-06	Brand Reputation Risk Monitoring	Detect digital-fatigue risks and potential brand backlash related to screen-time overuse.	M
BR-07	Revenue Opportunity Forecasting	Identify new revenue streams from digital wellness products, predictive models, and partnerships.	M
BR-08	Unified Data Integration Platform	Consolidate data from multiple sources (CDC, HR systems, wellness apps, surveys) into a single BI environment.	H
BR-09	Real-Time Dashboarding & Reporting	Provide interactive dashboards for HR teams, educators, and parents with real-time insights.	H
BR-10	Predictive & Prescriptive Analytics	Implement predictive models to forecast risks and prescriptive models to recommend actions.	H

BR-01: Screen Time Risk Identification

• **Business Context:**

Enables HR leaders, educators, and parents to understand which individuals or groups are at risk due to excessive screen time. In doing so, early intervention and long-term mental-health reduction is possible. As a result, adjustments can be implemented for optimum productivity impacts.

• **Data Sources:**

- Screen-time usage logs from wellness apps
- School survey data on daily device usage
- HR system data on employee device activity
- CDC research datasets on youth screen-time impacts

• **Business and Data Transformations**

Integrate data & apply risk scoring models.

• **Business Metrics or KPIs**

Screen time risk scores & at-risk group identification.

• **Business Processes**

HR wellness program management & intervention planning.

• **Business Groups Involved**

- HR team
 1. Handles risk communication to employees
 2. Reviews high-risk cases
 3. Coordinates early interventions
- Data engineers
 1. Integrate screen-time, HR, and survey datasets
 2. Build pipelines for risk scoring

3. Maintain data quality and ingestion

- EdTech partnership team

1. Coordinates data-sharing agreements
2. Aligns risk models with partner platforms
3. Supports integration with school systems

- **Business Owners**

- Jennifer Martinez, HR Director

- **Subject matter experts**

- Dr. Sarah Mitchell, Clinical Psychologist
- James Chen, Senior Data Scientist

BR-02: Productivity Impact Analysis

• **Business Context:**

Executives want to quantify how screen time affects productivity, absenteeism, and engagement. This supports workforce optimization and informs wellness program investments.

• **Data Sources**

- HR productivity metrics (uptime, task completion, absenteeism)
- Employee wellness app usage
- Time-tracking systems
- Screen-time logs

• **Business and Data Transformations**

Correlate screen time with productivity metrics & workforce optimization.

• **Business Metrics or KPIs**

Productivity impact scores & absenteeism rates.

• **Business Processes**

Workforce optimization & wellness program evaluation.

• **Business Groups Involved**

- HR team
 1. Provides productivity metrics
 2. Interprets workforce impact
 3. Implements policy changes
- Data analysts
 1. Correlate screen time with productivity
 2. Build dashboards and reports
 3. Validate statistical significance

- Executives

1. Review insights for decision-making
2. Approve wellness investments
3. Set productivity strategy

- **Business Owners**

- Michael Roberts, HR Director

- **Subject matter experts**

- Maria Rodriguez, Workforce Analytics Manager
- Dr. Kevin Park, Industrial-Organizational Psychologist

BR-03: Digital Wellness Intervention Recommendations

- **Business Context:**

Business owners need actionable recommendations—what interventions to apply, to whom, and when—to reduce digital fatigue and improve wellness outcomes.

- **Data Sources:**

- Predictive model outputs
- Historical intervention outcomes
- Screen-time patterns
- Mental-health survey responses

- **Business and Data Transformations**

Develop recommendation algorithms & intervention scoring models.

- **Business Metrics or KPIs**

Intervention effectiveness rates & employee engagement scores.

- **Business Processes**

Wellness program development & intervention planning.

- **Business Groups Involved**

- HR team
 1. Identifies employees needing interventions
 2. Communicates recommendations
 3. Tracks participation
- Wellness program managers
 1. Designs intervention programs
 2. Oversees rollout and scheduling
 3. Measures program effectiveness
- Data scientists

1. Build recommendation algorithms
2. Score intervention effectiveness
3. Optimize targeting logic

- **Business Owners**

- Sarah Johnson, Wellness Program Manager

- **Subject matter experts**

- Dr. Amanda Foster, Behavioral Health Specialist
 - Robert Thompson, Machine Learning Engineer

BR-04: EdTech & Wellness Partnership Enablement

• **Business Context:**

Partnership teams need analytics that demonstrate value to EdTech and wellness partners, enabling co-development of digital wellness tools.

• **Data Sources:**

- EdTech platform usage data
- Partner product performance metrics
- School and HR program adoption data

• **Business and Data Transformations**

Analyze partnership value & identify co-development opportunities.

• **Business Metrics or KPIs**

Partnership ROI & co-development metrics.

• **Business Processes**

Partnership development & business development.

• **Business Groups Involved**

- EdTech partnership team
 1. Manages partner relationships
 2. Aligns analytics with partner needs
 3. Coordinates co-development
- Business development team
 1. Identifies partnership opportunities
 2. Negotiates agreements
 3. Supports joint product strategy

- **Business Owners**

- David Chen, EdTech Partnership Manager

- **Subject matter experts**

- Jennifer Liu, Strategic Partnerships Director
- Michael Stevens, Product Analytics Lead

BR-05: Mental Health Trend Monitoring

• **Business Context:**

Mental-health officers and HR wellness teams need visibility into trends that correlate screen time with depression, anxiety, and burnout.

• **Data Sources:**

- CDC mental-health datasets
- Employee assistance program (EAP) anonymized data
- School mental-health survey results

• **Business and Data Transformations**

Analyze mental health trends & correlate with screen time patterns.

• **Business Metrics or KPIs**

Mental health trend indicators & screen time correlation scores.

• **Business Processes**

Mental health support services & wellness program evaluation.

• **Business Groups Involved**

- HR team
 1. Monitors employee well-being trends
 2. Coordinates support services
 3. Communicates findings to leadership
- Mental health professionals
 1. Interpret mental-health indicators
 2. Validate risk correlations
 3. Recommend support interventions
- Data analysts

1. Analyze mental-health datasets
2. Build trend visualizations
3. Identify emerging risk patterns

- **Business Owners**

- Dr. Sandra Sanderbelt, Mental Health Officer

- **Subject matter experts**

- Dr. Lisa Patel, Licensed Clinical Social Worker
 - David Kim, Epidemiologist

BR-06: Brand Reputation Risk Monitoring

- **Business Context:**

Marketing and PR teams need early warning indicators of digital-fatigue backlash or negative sentiment tied to screen-time overuse.

- **Data Sources:**

- Social media sentiment feeds
- Customer feedback surveys
- App store reviews
- Brand reputation monitoring tools

- **Business and Data Transformations**

Analyze sentiment & identify reputation risks.

- **Business Metrics or KPIs**

Reputation risk scores & sentiment analysis metrics.

- **Business Processes**

Brand management & PR management.

- **Business Groups Involved**

- Marketing team
 1. Monitors brand sentiment
 2. Responds to digital-fatigue trends
 3. Adjusts messaging strategy
- PR team
 1. Manages public communications
 2. Handles negative sentiment events
 3. Develops mitigation plans

- Data analysts

1. Run sentiment analysis models
2. Detect early warning signals
3. Produce reputation dashboards

- **Business Owners**

- Rebecca Williams, Marketing Director

- **Subject matter experts**

- Nicole Anderson, Social Media Analytics Manager
- Brian Walsh, NLP Specialist

BR-07: Revenue Opportunity Forecasting

• **Business Context:**

Executives need to understand where new revenue streams can emerge—digital wellness products, predictive analytics, EdTech partnerships.

• **Data Sources:**

- Market trend reports
- Product adoption analytics
- Sales pipeline data
- Competitive intelligence

• **Business and Data Transformations**

Analyze market trends & forecast revenue opportunities.

• **Business Metrics or KPIs**

Revenue opportunity forecasts & market trend indicators.

• **Business Processes**

Business development & revenue forecasting.

• **Business Groups Involved**

- Executives
 1. Define strategic revenue priorities
 2. Approve new product directions
 3. Evaluate opportunity forecasts
- Business development team
 1. Identifies new markets
 2. Validates revenue opportunities
 3. Supports go-to-market planning

- Data analysts

1. Analyze market trends
2. Build forecasting models
3. Quantify opportunity size

- **Business Owners**

- Tinkerbella Tamesia, Executive Director

- **Subject matter experts**

- Patricia Johnson, Financial Analyst
- Thomas Wright, Business Intelligence Manager

BR-08: Unified Data Integration Platform

• **Business Context:**

Data engineers and BI teams require a single, governed environment to integrate and manage all data sources for consistent analytics.

• **Data Sources:**

- CDC datasets
- HRIS systems
- Wellness apps
- EdTech platforms
- Internal databases

• **Business and Data Transformations**

Integrate data sources & develop data governance framework.

• **Business Metrics or KPIs**

Data integration metrics & data quality scores.

• **Business Processes**

Data management & data governance.

• **Business Groups Involved**

- Data engineers
 1. Build the unified data platform
 2. Manage pipelines and ETL
 3. Enforce data governance
- Data analysts
 1. Define analytical requirements
 2. Validate integrated datasets

3. Build BI layers

- IT team

- 1. Manage infrastructure

- 2. Oversee security and access

- 3. Support platform reliability

- **Business Owners**

- Capo Lucifer, Data Director

- **Subject matter experts**

- Christopher Lee, Data Architect

- Rachel Green, Data Governance Specialist

BR-09: Real-Time Dashboarding & Reporting

- **Business Context:**

HR teams, educators, and parents need intuitive dashboards that provide real-time insights into screen time, risks, and recommended actions.

- **Data Sources:**

- Live app usage feeds
- HR system updates
- Survey data
- Predictive model outputs

- **Business and Data Transformations**

Develop real-time dashboards & automate reporting.

- **Business Metrics or KPIs**

Dashboard adoption rates & report usage metrics.

- **Business Processes**

Business intelligence & data visualization.

- **Business Groups Involved**

- Data analysts
 1. Build dashboards
 2. Maintain real-time data flows
 3. Validate metrics
- HR team
 1. Uses dashboards for decision-making
 2. Provides requirements and feedback
 3. Communicates insights to stakeholders

- Executives

1. Review dashboards for strategic decisions
2. Track KPIs
3. Request enhancements

- **Business Owners**

- Kathy Junebug, BI Director

- **Subject matter experts**

- Angela Martinez, Data Visualization Expert
- Steven Cooper, UX Designer

BR-10: Predictive & Prescriptive Analytics

- **Business Context:**

Decision-makers want models that forecast risk and recommend specific actions to improve wellness and productivity.

- **Data Sources:**

- Historical screen-time data
- Intervention outcomes
- Mental-health indicators
- Productivity metrics

- **Business and Data Transformations**

Develop predictive models & prescriptive algorithms.

- **Business Metrics or KPIs**

Model accuracy scores & intervention effectiveness rates.

- **Business Processes**

Predictive analytics & prescriptive analytics.

- **Business Groups Involved**

- Data scientists
 1. Build predictive models
 2. Develop prescriptive algorithms
 3. Validate accuracy and fairness
- HR team
 1. Applies model outputs to workforce decisions
 2. Communicates recommendations
 3. Tracks outcomes
- Executives

1. Use predictions for strategic planning
2. Approve model-driven initiatives
3. Evaluate ROI

- **Business Owners**

- Canny Jenkins, Data Science Director

- **Subject matter experts**

- Dr. Samuel Brooks, Statistician
 - Emily Zhang, AI/ML Engineer
-

DATA AND DATA QUALITY REQUIREMENTS

Data sources

1. Screen-time usage logs from wellness apps

- Status: New source
- Data Quality: Needs assessment
- Owners: Wellness app providers, IT team
- SMEs: Data engineers

2. School survey data on daily device usage

- Status: Existing source, needs more data extracted
- Data Quality: Variable, needs cleansing
- Owners: School administrators, IT team
- SMEs: Data analysts

3. HR system data on employee device activity

- Status: Existing source
- Data Quality: Good
- Owners: HR department, IT team
- SMEs: HR data analysts

4. CDC research datasets on youth screen-time impacts

- Status: Existing external source
- Data Quality: High
- Owners: CDC
- SMEs: Data scientists

5. HR productivity metrics

- Status: Existing source
- Data Quality: Good
- Owners: HR department
- SMEs: HR data analystsData Cleansing
- Identify inconsistencies in screen-time data formats across sources
- Handle missing values in survey data
- Standardize device activity metrics from HR systems and wellness apps
- Validate CDC dataset integrity

Data cleansing

To ensure the data sources can be reliably integrated and used for analytics, several cleansing and standardization activities are required. These activities address inconsistencies across systems, missing or incomplete values, and variations in data definitions that prevent accurate reporting and modeling.

1. Standardize Screen-Time Data Across Sources

- Normalize timestamp formats from wellness apps, HR systems, and school surveys
- Align units of measurement (minutes vs. hours)
- Standardize device categories (mobile, tablet, laptop, shared device)
- Resolve discrepancies in session-logging logic across platforms

2. Clean and Validate School Survey Data

- Identify and handle missing or incomplete survey responses
- Remove duplicate or contradictory entries
- Standardize question formats and response scales
- Validate age, grade level, and demographic fields for consistency

3. Harmonize HR Device Activity Metrics

- Align HR system activity logs with wellness app usage definitions
- Standardize activity thresholds (active, idle, offline)
- Validate employee identifiers across HRIS and device-tracking systems
- Correct inconsistent or outdated department/role mappings

4. Integrate External CDC Datasets

- Validate dataset integrity and confirm version alignment
- Map CDC variable definitions to internal data model
- Standardize demographic categories (age groups, gender, region)
- Ensure CDC time-series data aligns with internal reporting periods

5. Resolve Cross-Source Data Conflicts

- Identify conflicting screen-time totals across systems
- Prioritize authoritative sources for overlapping fields
- Apply reconciliation rules for inconsistent timestamps
- Establish a unified definition of “screen-time session”

6. Apply Data Quality Checks

- Completeness checks for required fields (usage hours, productivity metrics, survey responses)
- Accuracy checks against known ranges (e.g., screen time cannot exceed 24 hours/day)
- Consistency checks across related fields (e.g., device activity vs. productivity logs)
- Outlier detection for anomalous usage patterns

7. Prepare Data for Modeling and BI

- Convert raw logs into aggregated daily/weekly metrics
- Create standardized keys for joining datasets (student ID, employee ID, device ID)
- Apply data governance rules for sensitive fields
- Ensure all datasets meet minimum quality thresholds before integration

Master Data Management (MDM)

- No specific MDM requirements identified for customer or product data
- Potential need for data governance framework for screen-time and wellness data

Data acquisition

- Obtain external data on digital wellness trends if needed
- Potential need to acquire additional screen-time data from other sources (e.g., mobile carriers)

FUNCTIONAL REQUIREMENTS

BI use cases

UC-01: *Identify At-Risk Groups*

Users analyze screen-time patterns to detect individuals or groups at risk for mental-health or productivity decline.

UC-02: *Monitor Productivity & Wellness Trends*

Users explore dashboards showing correlations between screen time, productivity, absenteeism, and wellness.

UC-03: *Generate Personalized Intervention Plans*

Predictive models recommend targeted interventions for employees, students, or children.

UC-04: *Support EdTech & Wellness Partnerships*

Analytics demonstrate value to partners and guide co-development of digital wellness tools.

UC-05: *Forecast Revenue Opportunities*

Executives use BI tools to evaluate potential revenue streams from digital wellness solutions.

Analytical process workflow and user interaction

UC-01: *Identify At-Risk Groups*

- User selects population (employees, students, children)
- Dashboard displays screen-time distribution
- User filters by age, role, department, or school
- Predictive model flags high-risk individuals
- User exports list for intervention planning

UC-02: *Monitor Productivity & Wellness Trends*

- User views productivity KPIs
- Drill-down into screen-time correlations
- Compare trends across time periods
- Identify anomalies or spikes
- Generate automated insights

UC-03: *Generate Personalized Intervention Plans*

- User reviews model-generated recommendations
- Selects intervention type (break reminders, app limits, wellness coaching)
- Tracks outcomes over time

UC-04: *Support EdTech & Wellness Partnerships*

- User views partner-specific dashboards
- Analyzes adoption, engagement, and outcomes
- Generates reports for partner meetings

UC-05: *Forecast Revenue Opportunities*

- User reviews market trends
- Applies scenario modeling
- Forecasts revenue impact of new wellness products

Analytical styles needed

Descriptive Analytics

Used across all use cases for dashboards, KPIs, and trend analysis.

Diagnostic Analytics

Used by analysts and HR teams to drill into root causes of productivity or wellness issues.

Predictive Analytics

Used in UC-01, UC-03, UC-05 to forecast risks and revenue opportunities.

Prescriptive Analytics

Used in UC-03 to recommend interventions.

Self-Service BI

For power users (HR analysts, educators) who build custom reports.

Operational BI

For real-time dashboards used by HR teams and parents.

Executive BI

For leadership reviewing high-level KPIs and revenue forecasts.

REGULATORY/COMPLIANCE REQUIREMENTS

FERPA (Family Educational Rights and Privacy Act)

- Applies to all student screen-time data collected from schools
- Requires written consent for data sharing and strict access controls
- Mandates secure storage and limited disclosure of educational records

COPPA (Children's Online Privacy Protection Act)

- Applies to data collection from children under 13
- Requires verifiable parental consent before collecting personal information
- Mandates data deletion procedures and privacy policy disclosures

HIPAA (Health Insurance Portability and Accountability Act)

- May apply if mental health data is collected through Employee Assistance Programs (EAP)
- Requires Business Associate Agreements (BAAs) with data providers
- Mandates encryption, audit logs, and breach notification procedures

GDPR/State Privacy Laws (CCPA, VCDPA, etc.)

- Applies if collecting data from EU residents or certain U.S. states
- Requires data minimization, purpose limitation, and right to deletion
- Mandates privacy impact assessments and data protection by design

Data Retention and Destruction Policies

- Must establish retention schedules for all collected data
- Required secure deletion procedures after retention period expires
- Compliance with organizational and legal record-keeping requirements

IRB (Institutional Review Board) Approval

- Required if research involves minors or vulnerable populations
- Mandates ethical review of data collection methods and consent procedures
- Ongoing monitoring and annual renewal requirements

TECHNICAL REQUIREMENTS

Table 2: Technical Requirements

Requirement ID	Technical Requirement Description	Impacted Business Requirements	Options for Compliance
TR-01	Hybrid Cloud Integration: Capability to securely ingest data from internal HRIS (on-prem) and external Wellness/CDC APIs (cloud).	BR-08, BR-10	Azure Data Factory, AWS Glue, or Talend Open Studio.
TR-02	PII Masking & Encryption: Automated masking of Personal Identifiable Information (PII) for student and employee IDs during ingestion.	BR-01, BR-05	AES-256 encryption at rest; Role-Based Access Control (RBAC).
TR-03	Real-Time Stream Processing: Infrastructure to support sub-minute latency for live app usage feeds.	BR-09	Apache Kafka or AWS Kinesis for stream handling.
TR-04	ML Model Deployment Environment: A dedicated environment to host Python/R-based predictive models (Risk Scoring).	BR-03, BR-10	Docker containers managed via Kubernetes or SageMaker.
TR-05	Mobile-Responsive UI: Dashboards must be accessible via mobile browsers for parents and traveling executives.	BR-09	Power BI Mobile or Tableau Mobile optimization.
TR-06	High Availability (99.9%): The BI platform must be available during school/business hours without downtime.	BR-09	Multi-availability zone deployment.

REQUIREMENTS CROSS-MATRIX

- **Strategic Value Dimension**

High Strategic Value:

- BR-01: Screen Time Risk Identification
- BR-08: Unified Data Integration Platform
- BR-09: Real-Time Dashboarding & Reporting
- BR-10: Predictive & Prescriptive Analytics

Medium Strategic Value:

- BR-02: Productivity Impact Analysis
- BR-03: Digital Wellness Intervention Recommendations
- BR-05: Mental Health Trend Monitoring

Low Strategic Value:

- BR-04: EdTech & Wellness Partnership Enablement
- BR-06: Brand Reputation Risk Monitoring
- BR-07: Revenue Opportunity Forecasting

- **Implementation Complexity Dimension**

High Complexity:

- BR-08: Unified Data Integration Platform (multiple source integration)
- BR-10: Predictive & Prescriptive Analytics (ML model development)
- TR-03: Real-Time Stream Processing

Medium Complexity:

- BR-01: Screen Time Risk Identification
- BR-09: Real-Time Dashboarding & Reporting
- TR-04: ML Model Deployment Environment

Low Complexity:

- BR-02: Productivity Impact Analysis
- BR-06: Brand Reputation Risk Monitoring

- **Data Quality Impact Dimension**

Critical Data Quality Dependency:

- BR-01: Screen Time Risk Identification
- BR-10: Predictive & Prescriptive Analytics
- BR-05: Mental Health Trend Monitoring

Moderate Data Quality Dependency:

- BR-02: Productivity Impact Analysis
- BR-03: Digital Wellness Intervention Recommendations

Low Data Quality Dependency:

- BR-07: Revenue Opportunity Forecasting
- BR-06: Brand Reputation Risk Monitoring

- **Regulatory Risk Dimension**

High Regulatory Risk:

- BR-01: Screen Time Risk Identification (FERPA, COPPA)
- BR-05: Mental Health Trend Monitoring (HIPAA)
- BR-08: Unified Data Integration Platform (All compliance)

Medium Regulatory Risk:

- BR-02: Productivity Impact Analysis (Employee privacy)
- BR-09: Real-Time Dashboarding (Data access controls)

Low Regulatory Risk:

- BR-06: Brand Reputation Risk Monitoring
- BR-07: Revenue Opportunity Forecasting

- **Business Impact (ROI) Dimension**

High ROI Potential:

- BR-01: Screen Time Risk Identification (early intervention cost savings)
- BR-02: Productivity Impact Analysis (workforce optimization)
- BR-03: Digital Wellness Intervention Recommendations (reduced absenteeism)

Medium ROI Potential:

- BR-07: Revenue Opportunity Forecasting (new revenue streams)
- BR-04: EdTech & Wellness Partnership Enablement

Low/Indirect ROI:

- BR-06: Brand Reputation Risk Monitoring (risk mitigation)
- BR-05: Mental Health Trend Monitoring (long-term benefits)

PRIORITIZING REQUIREMENTS

Table 3: Prioritizing Requirements

Identifier	Requirement Name	Priority	Business Owner
BR-08	Unified Data Integration Platform	H	Capo Lucifer (Data Director)
BR-01	Screen Time Risk Identification	H	Jennifer Martinez (HR Director)
BR-05	Mental Health Trend Monitoring	H	Dr. Sandra Sanderbelt (Mental Health Officer)
BR-09	Real-Time Dashboarding & Reporting	H	Kathy Junebug (BI Director)
BR-02	Productivity Impact Analysis	M	Michael Roberts (HR Director)
BR-10	Predictive & Prescriptive Analytics	M	Canny Jenkins (Data Science Director)
BR-03	Digital Wellness Intervention Recommendations	M	Sarah Johnson (Wellness Manager)
BR-04	EdTech & Wellness Partnership Enablement	L	David Chen (Partnership Manager)
BR-06	Brand Reputation Risk Monitoring	L	Rebecca Williams (Marketing Dir)
BR-07	Revenue Opportunity Forecasting	L	Tinkerbelle Tamesia (Exec Director)

APPROVALS

Title	Name	Approval Date
Executive Sponsor	Dr. Melissa Carter	
Data Sponsor	James Patel	
BI Director	Kathy Junebug	
Director of HR	Jennifer Martinez	
Project Manager	Sarah Nguyen	

APPENDIX: INPUTS FOR REQUIREMENTS

Interviewees

- **Interviewees**

Jennifer Martinez, HR Director

- Title: Director of Human Resources
- Organizational Group: HR Department
- Project Role: Business Owner (BR-01)
- Involvement: Multiple sessions on productivity metrics, wellness program requirements, and risk identification needs. Key stakeholder for workforce-related analytics.

Dr. Sandra Sanderbelt, Mental Health Officer

- Title: Chief Mental Health Officer
- Organizational Group: Employee Wellness
- Project Role: Business Owner (BR-05), SME
- Involvement: Extensive consultation on mental health indicators, ethical considerations for data collection, and interpretation of behavioral patterns.

Capo Lucifer, Data Director

- Title: Director of Data and Analytics
- Organizational Group: IT/Data Department
- Project Role: Business Owner (BR-08), Technical Lead
- Dates: 1/15/26, 1/22/26, 2/5/26 - Requirements workshops
- Involvement: Data integration architecture, governance framework design, and technical feasibility assessments.

Kathy Junebug, BI Director

- Title: Director of Business Intelligence
- Organizational Group: Analytics Department
- Project Role: Business Owner (BR-09), Project Lead
- Involvement: Continuous involvement throughout requirements gathering. Led dashboard design sessions and BI platform selection.

Canny Jenkins, Data Science Director

- Title: Director of Data Science
- Organizational Group: Analytics Department
- Project Role: Business Owner (BR-10), ML Lead
- Dates: 1/18/26, 1/25/26, 2/8/26
- Involvement: Predictive modeling requirements, algorithm selection, and model deployment architecture.

David Chen, EdTech Partnership Manager

- Title: EdTech Partnership Manager
- Organizational Group: Business Development
- Project Role: Business Owner (BR-04)
- Dates: 1/20/26, 2/3/26
- Involvement: Partnership analytics requirements and co-development opportunities.

Dr. Sarah Mitchell, Clinical Psychologist

- Title: Clinical Psychologist
- Organizational Group: External Consultant
- Project Role: Subject Matter Expert (BR-01)
- Dates: 1/12/26, 1/30/26
- Involvement: Risk assessment methodologies and ethical guidelines for youth data.

James Chen, Senior Data Scientist

- Title: Senior Data Scientist
- Organizational Group: Data Science Team
- Project Role: Subject Matter Expert (BR-01, BR-10)
- Involvement: Model development requirements and technical validation.

School District Representatives

- Multiple school administrators from partner districts
- Dates: 1/10/26 - 1/17/26 (group sessions)
- Involvement: Data availability discussions, privacy requirements, and use case validation.

Data Sources Examined

Internal HRIS (Workday)

- Type: Relational Database (Oracle)
- Owner: HR Department
- Status: Examined for productivity metrics, employee demographics, attendance records
- Quality Assessment: High data quality, well-maintained
- Relevance: Primary source for BR-02, BR-03

Employee Wellness App Database (Virgin Pulse)

- Type: Cloud API
- Owner: Wellness Program Team
- Status: Examined for screen-time logs, activity tracking, engagement metrics
- Quality Assessment: Variable quality, some missing timestamps
- Relevance: Critical for BR-01, BR-02, BR-05

School District Student Information Systems (PowerSchool)

- Type: External Database (MySQL)
- Owner: Partner School Districts
- Status: Sample data examined under data-sharing pilot agreement
- Quality Assessment: Inconsistent formats across districts, needs standardization
- Relevance: Essential for BR-01, youth-focused analytics

CDC Youth Risk Behavior Survey (YRBS) Datasets

- Type: Public Dataset (CSV files)
- Owner: Centers for Disease Control and Prevention
- Status: Downloaded and examined for baseline benchmarking
- Quality Assessment: High quality, comprehensive documentation
- Relevance: Reference data for BR-05

Internal Time-Tracking System (Toggl)

- Type: Cloud SaaS
- Owner: IT Department
- Status: Examined for device activity correlation with productivity
- Quality Assessment: Good quality, real-time updates
- Relevance: Supporting data for BR-02

EdTech Platform Analytics (Canvas LMS)

- Type: Cloud API
- Owner: Partner EdTech Providers
- Status: Sample data examined
- Quality Assessment: Good structure, needs mapping to internal IDs
- Relevance: BR-04 partnership analytics

Social Media Sentiment Data (Brandwatch)

- Type: Third-party API
- Owner: Marketing Department subscription
- Status: Examined for brand reputation monitoring feasibility
- Quality Assessment: High volume, requires NLP processing
- Relevance: BR-06

Sales CRM (Salesforce)

- Type: Cloud Database
- Owner: Sales Department
- Status: Examined for partnership and revenue opportunity tracking
- Quality Assessment: Good quality, well-structured
- Relevance: BR-07

Data Source Systems Documentation

Workday HRIS Documentation

- Complete data dictionary and entity-relationship diagrams reviewed
- API documentation for automated extraction examined
- Security and access control policies documented

Virgin Pulse API Documentation

- REST API specifications reviewed
- Data schema and available endpoints documented
- Rate limiting and authentication requirements noted

PowerSchool Integration Guide

- FERPA compliance documentation reviewed
- Data extraction procedures and file formats examined
- Student identifier mapping protocols documented

CDC YRBS Codebook

- Variable definitions and survey methodology reviewed
- Sampling weights and statistical guidance examined
- Data use restrictions and citation requirements documented

Tableau Server Architecture Guide

- Deployment options and scalability reviewed
- Security features and row-level security capabilities examined
- Mobile responsiveness capabilities validated

Reporting Systems, Reports and Shadow Systems Examined

Existing HR Dashboard (Excel-based)

- Current monthly productivity reports reviewed
- Manual reporting process documented (12 hours/month)
- Identified opportunity for automation (BR-09)

School Wellness Reports (PDF)

- Annual wellness survey results from partner schools examined
- Manual aggregation process identified as inefficient
- Data exists in various formats requiring standardization

Employee Assistance Program (EAP) Reports

- Anonymized mental health utilization reports reviewed
- Quarterly trends in counseling requests examined
- Identified correlation opportunities with screen-time data

Marketing Dashboard (Google Analytics)

- Brand sentiment tracking reviewed
- Limited integration with operational data identified
- Opportunity for unified view (BR-06)

Shadow System: HR Analyst Personal Database

- Discovered Access database maintained by HR analyst
- Contains manually compiled productivity correlations
- Highlights need for unified platform (BR-08)

Shadow System: School Counselor Spreadsheets

- Multiple Excel files tracking student wellness concerns
- Inconsistent formats and manual updates
- Critical data source requiring integration

Executive Dashboard (Power BI)

- Current executive KPI dashboard reviewed
- Limited drill-down capabilities identified
- Requirements gathered for enhancement (BR-09)

Authors and contributors

- **Hali Sarah Parsons (Lead Author):** Primary BI Strategist and document architect. Responsible for the Business Justification and Technical Requirements alignment.
- **Dr. Melissa Carter (Executive Sponsor):** Provided strategic vision for youth employability and long-term workforce impact metrics.
- **James Patel (Data Sponsor):** Provided data governance standards and access to the Youth Digital Wellness Coalition datasets.
- **Jennifer Martinez (HR Director):** Contributed "at-risk" behavioral indicators and employee productivity KPI definitions.
- **Dr. Amanda Foster (SME):** Clinical Psychologist who provided the logic for the mental health risk scoring models.
- **Christopher Lee (SME):** Data Architect who designed the cloud-native integration paths for the CDC and wellness app feeds.